

ARYAN KUMAR SINHA

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Summary

M.Tech candidate in AI ML with hands-on experience building NLP pipelines, RAG systems, and LLM-integrated applications. Proven ability to work across the full GenAI stack from prompt design and embedding-based retrieval to agentic workflows and production API integration. Seeking to apply applied GenAI engineering skills within an enterprise innovation context.

EXPERIENCE

La Trobe University

Oct. 2023 – Apr. 2024

Research Intern

- Engineered a real-time smishing detection system using ANN models and NLP-based feature pipelines including TF-IDF vectorization and word embeddings achieving **98% classification accuracy** across 50,000+ SMS samples
- Reduced threat analysis time by **30%** by optimizing model inference pipelines, cutting false-negative rates and improving detection speed by **40%** in a production-deployed mobile application
- Designed and evaluated prompt-style classification strategies and 6 feature engineering experiments to iteratively improve model performance, applying a structured experimentation and evaluation workflow across the full ML lifecycle
- Built and shipped a user-facing mobile application integrating the detection model via REST API, serving real-time predictions to end users with **sub-400ms** response time

SKILLS

Languages: Java, Python, C, SQL, Bash

Frameworks & Libraries: JavaFx, Spring Boot, Flask, cProfile, OpenCV, SAM, TensorFlow, Scikit-learn, Pandas, Selenium, Playwright

Web & APIs: REST APIs, HTML, CSS, JavaScript, OpenAI API, Hugging Face Inference API, Anthropic API

Databases: MySQL, MongoDB, FAISS, ChromaDB

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, AI Agents / Agentic AI, RAG, Feature Engineering, Prompt Engineering, LangChain, LangGraph, Vector Databases, Embeddings, Function / Tool Calling, Generative AI APIs

Tools & Platforms: Git, GitHub, GitHub Actions (CI/CD), Docker, AWS, Prometheus, Linux, Postman, VS Code, Power BI

Soft Skills: Communication, Team Collaboration, Critical Thinking, Adaptability, Time Management, Quick Learner

PROJECTS

SG Place Watch - Automated POI Change Detection System

2026

- Built a geospatial system to detect new and closed businesses across **10 POI categories** using OpenStreetMap as the baseline
- Integrated **15+ data sources (6 government registries, 9 commercial platforms)** to validate real-world business activity
- Developed a multilingual NLP pipeline supporting **4 languages** for event classification, translation, semantic entity matching, and embedding-based similarity search
- Engineered a RAG-based stateful closure-detection workflow with SQLite-backed history tracking, retrieval-augmented validation, explainability reports, GeoJSON exports, and a Flask dashboard for reviewing predictions
- Processed **3,500+ locations** per run and surfaced **340+ high-confidence place updates** to OpenStreetMap dataset

DreamSketch EEG-to-Image Generative AI Pipeline

2026

- Developed and trained an EEG encoder to map sleep EEG signals into semantic embedding representations for generative AI image synthesis
- Preprocessed **2,000+ EEG-dream report pairs** using bandpass filtering, STFT spectrograms, and occipital-channel extraction, creating model-ready neural representations and achieving **0.69 cosine similarity** and **22% Top-10 retrieval accuracy** on unseen samples

Multi-Modal Earnings Call Volatility Prediction

2026

- Built a multi-modal pipeline combining earnings call transcripts, audio sentiment, and historical stock data for volatility prediction
- Processed and analyzed **2k+ earnings calls** using NLP techniques, audio feature extraction, and financial time-series preprocessing
- Trained and evaluated multiple LLM and ML models including Praat, FinBERT, wav2vec2, and ensemble methods
- Achieved **Spearman $\rho = 0.42$** on held-out earnings calls, a **35% improvement** over text-only models and accurately ranking post-earnings stock volatility through multimodal fusion

EDUCATION

Master of Technology in Artificial Intelligence & Machine Learning

Aug. 2025 – May 2027

Symbiosis Institute of Technology, Pune, MH

GPA: 7.15

Bachelor of Technology in Computer Science & Engineering

Aug. 2020 – May 2024

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai, TN

CGPA: 8.57

CERTIFICATIONS

AI Agents in LangGraph DeepLearning.AI x LangChain	2026
Data Engineering using Databricks on AWS and Azure Udemy	2025
Apache Kafka Series - Kafka Streams for Data Processing Udemy	2025
Generative AI Engineering with LLMs Specialization IBM x Coursera	2025
Data Analysis for Biologists NPTEL	2023